Physics P1: Solutions

Physical Reasoning and Estimation · MAIS PAC

Problem 1.

(a) Write $\mathbf{v} = \mathbf{C}\,\lambda^{\mathbf{a}}\,\mathbf{g}^{\mathbf{b}}\,\rho^{\mathbf{c}}$. Since $[\mathbf{v}] = [\mathbf{L}][\mathbf{T}]^{-1}$, $[\lambda] = [\mathbf{L}]$, $[\mathbf{g}] = [\mathbf{L}][\mathbf{T}]^{-2}$, and $[\rho] = [\mathbf{M}][\mathbf{L}]^{-3}$, matching exponents of $[\mathbf{M}]$ immediately gives $\mathbf{c} = 0$: ρ is the **only** parameter carrying mass, so nothing can cancel it! (Physically, doubling water's density doubles both the gravitational restoring weight and the inertia of each fluid parcel, so the two cancel.) Matching $[\mathbf{T}]$ gives $-2\mathbf{b} = -1 \Rightarrow \mathbf{b} = 1/2$, and matching $[\mathbf{L}]$ gives $\mathbf{a} + \mathbf{b} = 1 \Rightarrow \mathbf{a} = 1/2$. Thus

$$\mathbf{v} = \mathbf{C}\,\sqrt{(\mathbf{g}\lambda)} \tag{1}$$

(exact fluid theory gives $\mathbf{C} = 1/\sqrt{(2\pi)} \approx 0.40$, so longer ocean swells outrun shorter chop).

(b) When $\mathbf{h} \ll \lambda$ and $\mathbf{v}$ is independent of λ, the same reasoning with $\mathbf{h}$ replacing λ gives $\mathbf{v} = \mathbf{C}'\,\sqrt{(\mathbf{gh})}$ (in fact $\mathbf{C}' = 1$ exactly for linear shallow-water waves). For $\mathbf{h} = 4.0 \times 10^3$ m,

$$\mathbf{v} \approx \sqrt{((9.8\ \text{m/s}^2)(4000\ \text{m}))} \approx 198\ \text{m/s} \approx 710\ \text{km/h}, \tag{2}$$

comparable to a commercial jetliner.

Problem 2.

(a) Test each limit in turn:

1. **Flat wedge** ($\theta \to 0$): the block sits at rest on a flat horizontal surface, so we must have $\mathbf{A} \to 0$. Candidate (i) gives $\mathbf{mg}/(\mathbf{M} + \mathbf{m}) \neq 0$, eliminating **(i)**.
2. **Vertical wall** ($\theta \to \pi/2$): the block drops straight down beside a vertical cliff without pressing horizontally against the wedge, so again $\mathbf{A} \to 0$. Candidate (iv) diverges ($\tan\theta \to \infty$), eliminating **(iv)**.
3. **Immovable wedge** ($\mathbf{M} \to \infty$): an infinitely massive wedge cannot accelerate, so $\mathbf{A} \to 0$. Candidate (ii) approaches $\mathbf{g}\sin\theta\cos\theta \neq 0$ as $\mathbf{M} \to \infty$, eliminating **(ii)**.
4. **Heavy wedge on a gentle slope** ($\mathbf{M} \gg \mathbf{m}$ and small θ, with $\mathbf{M}\theta^2 \ll \mathbf{m}$): the block presses on the nearly flat wedge with normal force $\mathbf{N} \approx \mathbf{mg}$, exerting a tiny horizontal force $\mathbf{N}\sin\theta \approx \mathbf{mg}\theta$ on a massive wedge of mass $\mathbf{M}$, so Newton's second law requires $\mathbf{A} \approx (\mathbf{m}/\mathbf{M})\mathbf{g}\theta$. Candidate (iii) gives $\mathbf{mg}\theta/\mathbf{M}$ as expected, whereas candidate (v) gives $\mathbf{mg}\theta/\mathbf{m} = \mathbf{g}\theta$ (independent of $\mathbf{M}$!), eliminating **(v)**. Only **(iii)** survives!

(b) Let the slope descend to the right and let the wedge recoil to the left with acceleration $\mathbf{A}$. If $\mathbf{N}$ is the normal force between block and wedge, horizontal motion of the wedge gives $\mathbf{MA} = \mathbf{N}\sin\theta$. Because the block remains in contact with the wedge, its acceleration component along the outward normal $(\sin\theta, \cos\theta)$ must equal that of the wedge, namely $-\mathbf{A}\sin\theta$. Newton's second law for the block along the normal gives

$$\mathbf{N} - \mathbf{mg}\cos\theta = -\mathbf{mA}\sin\theta \Rightarrow (\mathbf{MA})/(\sin\theta) = \mathbf{mg}\cos\theta - \mathbf{mA}\sin\theta,$$

(3)

which rearranges directly to $\mathbf{A} = (\mathbf{mg} \sin\theta \cos\theta)/(\mathbf{M} + \mathbf{m}\sin^2\theta)$, confirming **(iii)**.

Problem 3.

(a) Because the atmospheric scale height (≈ 8.5 km) is roughly 0.13% of $\mathbf{R}_\oplus$, gravitational acceleration $\mathbf{g}$ is essentially uniform across nearly the entire mass of the air, and the spherical shell has surface area $4\pi\mathbf{R}_\oplus^2$. Hydrostatic balance requires sea-level pressure to support the weight per unit area of the air column overhead: $\mathbf{P}_0 = \mathbf{M}_{atm}\mathbf{g} / (4\pi\mathbf{R}_\oplus^2)$. Hence

$$\mathbf{M}_{atm} \approx (4\pi\, \mathbf{R}_\oplus^2\, \mathbf{P}_0)/(\mathbf{g}) = (4\pi\,(6.37 \times 10^6\ \text{m})^2\,(1.01 \times 10^5\ \text{Pa}))/(9.8\ \text{m/s}^2) \approx 5.3 \times 10^{18}\ \text{kg}. \tag{4}$$

(Accounting for continental elevation lowers the true global value slightly to 5.15×10^{18} kg.)

(b) Local pressure measures the weight per unit area of the air **above** that altitude. Since $\mathbf{P}_{UB} / \mathbf{P}_0 \approx 8.6 \times 10^4 / 1.01 \times 10^5 \approx 0.85$, roughly 85% of an idealized sea-level air column lies above 1350 m, and about **15%** lies below that altitude. This is a local column estimate, not the fraction of the whole atmosphere below 1350 m, which depends on Earth's topography.

Problem 4.

(a) We have $[\mathbf{G}] = [\mathbf{L}]^3[\mathbf{M}]^{-1}[\mathbf{T}]^{-2}$, $[\rho_0] = [\mathbf{M}][\mathbf{L}]^{-3}$, and $[\mathbf{R}_0] = [\mathbf{L}]$. Notice that $[\mathbf{G}\rho_0] = [\mathbf{T}]^{-2}$ already cancels both mass and length! Any factor of $\mathbf{R}_0$ would introduce unmatched length dimensions, so

$$\mathbf{t}_{ff} = \mathbf{C}\,(1)/(\sqrt{(\mathbf{G}\rho_0)}). \tag{5}$$

Physically, if we double $\mathbf{R}_0$ at fixed ρ_0, each shell has twice as far to fall, **but** the enclosed mass $\mathbf{M} = (4)/(3)\pi\mathbf{R}_0^3\rho_0$ increases eightfold so surface gravity $\mathbf{GM}/\mathbf{R}_0^2 \propto \rho_0\mathbf{R}_0$ also doubles; from $\mathbf{R}_0 = (1)/(2)\mathbf{gt}^2$, the two factors of 2 cancel! (Exact integration gives $\mathbf{C} = \sqrt{(3\pi/32)} \approx 0.543$.)

(b) The Sun's mean density is $\bar{\rho}_\odot = \mathbf{M}_\odot / ((4)/(3)\pi\mathbf{R}_\odot^3) \approx 1.41 \times 10^3\ \text{kg/m}^3$. Taking $\mathbf{C} \approx 1$ gives $1/\sqrt{(\mathbf{G}\bar{\rho}_\odot)} \approx 3.26 \times 10^3$ s ≈ 54 minutes (or **29 minutes** with the exact prefactor $\sqrt{(3\pi/32)}$). The real Sun does not collapse in half an hour because nuclear fusion in its 1.5×10^7 K core maintains high thermal gas pressure that balances inward gravity at every radius (**hydrostatic equilibrium**).